

Haibo Zhao

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EDUCATION

Northeastern University

M.S. in Computer Science

Boston, MA **GPA: 3.9**

Sep. 2022 – May 2025

Xi'an Jiaotong University

B.Eng. in Electronics Science and Technology (National Scholarship)

Xi'an, China

Sep. 2016 – Jun. 2020

PUBLICATIONS

Generalizable Hierarchical Skill Learning via Object-Centric Representation, *RA-L (Under Review)*

Haibo Zhao, Yu Qi, Boce Hu, Yizhe Zhu, Ziyang Chen, Xupeng Zhu, Owen Howell, Haojie Huang, Robin Walters, Dian Wang*, Robert Platt* → [\[Project Page\]](#)

Proposed GSL, a hierarchical policy learning framework bridging vision-language models and visuomotor control via object-centric skills, achieving strong generalization and 15.5% higher performance using 10× less data.

BEAR: Benchmarking and Enhancing Multimodal Language Models for Atomic Embodied Capabilities, *ICLR (Under Review)*

Yu Qi*, **Haibo Zhao***, Ziyu Guo*, Siyuan Ma, Ziyang Chen, Yaokun Han, Renrui Zhang, Zitiantao Lin, Shiji Xin, Yijian Huang, Kai Cheng, Peiheng Wang, Jiazheng Liu, Jiayi Zhang, Yizhe Zhu, Wenqing Wang, Yiran Qin, Xupeng Zhu, Haojie Huang, Lawson L.S. Wong → [\[Project Page\]](#)

Proposed BEAR, a large-scale benchmark for atomic embodied tasks, and improved VLM agent performance through more effective tool invocation and multimodal reasoning.

Hierarchical Equivariant Policy via Frame Transfer, *ICML 2025*

Haibo Zhao*, Dian Wang*, Yizhe Zhu, Xupeng Zhu, Owen Howell, Linfeng Zhao, Yaoyao Qian, Robin Walters, Robert Platt → [\[Project Page\]](#)

Introduced a frame-transfer interface to enforce equivariance in hierarchical imitation learning, improving generalization and sample efficiency.

Equivariant Diffusion Policy, *CoRL 2024. Best Paper Finalist*

Dian Wang, Stephen Hart, David Surovik, Tarik Kelestemur, Haojie Huang, **Haibo Zhao**, Mark Yeatman, Jiuguang Wang, Robin Walters, Robert Platt → [\[Project Page\]](#)

Designed an SO(2)-equivariant diffusion policy for 6-DoF control with 21.9% average performance gains over baselines, validated in both simulation and real-world settings.

PROFESSIONAL EXPERIENCE

Research Assistant

Helping Hands Lab, NEU/ RAI (former Boston Dynamics AI)

Jun. 2024 – Jul. 2025

Boston, MA

Applied Scientist Co-op

Amazon Robotics

Jan. 2024 – Jun. 2024

Boston, MA

- Built a Kalman Filter based sharpness estimator to monitor camera focus status across 10+ Amazon workstations, improving health status.
- Designed and deployed a scalable health-monitoring system using AWS EC2, SNS, and SQS to ensure operational robustness.

Software Engineer Intern

Coinbase

May 2023 – Sep. 2023

San Francisco, CA

- Architected a bulk transfer pipeline in Go, increasing remediation throughput from 5K to over 1M concurrent requests.
- Integrated MongoDB, GraphQL, gRPC, Kafka, and AWS services (EC2, S3) to support scalable, paginated fund transfers.
- Built a React-based frontend to streamline user workflows and improve system accessibility.

Software Development Intern

AMD

Jun. 2022 – Sep. 2022

Back-end Development Intern

Bytedance

Apr. 2022 – Jun. 2022

Machine Learning Engineer Intern

iFLYTEK AI Lab

Apr. 2021 – Jun. 2021

SERVICE

Reviewer, **Conference on Robot Learning (CoRL)** 2025

Reviewer, **IEEE Robotics and Automation Letters (RA-L)** 2025

AWARDS

Northeastern University Research Apprenticeship, 2025